

DataDose



Intelligent Pharmaceutical Data Pipeline
& Real-Time Clinical Alert System

A Real-Time Clinical Intelligence Platform engineered to eliminate fatal medication errors and operationalize Personalized Medicine — combining Zero-Latency Kafka Streaming, Graph-Based Polypharmacy Analysis via Neo4j, and Hybrid GraphRAG AI.

Submitted to: Ministry of Communications and Information Technology
Digital Egypt Pioneers Initiative (DEPI) × eYouth Academy

01 Project Description

DataDose is not a conventional prescription-checking system. It is a Real-Time Clinical Intelligence Platform engineered to eliminate fatal medication errors and operationalize the paradigm of Personalized Medicine at institutional scale. By treating every medical prescription as a complex, interconnected network rather than a flat list of drug names, DataDose delivers a level of clinical intelligence previously unavailable to frontline healthcare practitioners.

THREE CORE TECHNICAL PILLARS

Zero-Latency Streaming

Apache Kafka deployed on Aiven Cloud captures prescription events from clinics and pharmacies and routes them through the processing pipeline within fractions of a second. Every event is published to a dedicated Kafka topic and consumed by downstream processors in real time, ensuring no critical interaction alert is ever delayed.

Polypharmacy Network Analysis

A Neo4j Aura Cloud graph database models the complete pharmacological universe — drugs, conditions, interactions, contraindications, and patient histories — as a rich property graph. This enables detection of direct pairwise drug interactions and cumulative N-degree polypharmacy risks invisible to traditional rule-based engines.

Hybrid AI — GraphRAG & LLMs

Groq-powered large language models cleanse and standardize free-text drug names with accuracy exceeding 90%. The GraphRAG architecture merges semantic LLM reasoning with structural graph traversal, enabling a natural-language Virtual Medical Assistant that answers complex, context-sensitive clinical queries in real time.

All processed data flows into a **Snowflake** cloud data warehouse structured on the Medallion Architecture, feeding live **Power BI** dashboards that give healthcare administrators a comprehensive, real-time view of prescription activity, interaction trends, and AI performance metrics.

02 Team Members & Roles

Youssef Adel

*Team Leader ·
Graph Database Designer ·
Full-Stack Developer*

Directed overall project planning, architecture decisions, and cross-functional coordination. Designed and implemented the Neo4j graph database schema including drug relationships, ATC classifications, and interaction pathways. Oversaw clinical data collection and curation, and led development of the full-stack web platform encompassing backend API design and frontend integration.

Mohamed Salah

*Data Engineer · Business
Intelligence Developer*

Developed and managed the Apache Kafka streaming infrastructure including topic configuration, producer/consumer pipelines, and fault-tolerance tuning. Contributed to Apache Airflow DAG design for workflow orchestration. Built Power BI dashboards for analytical reporting and KPI monitoring.

Amr Ayman

*Cloud Data Engineer ·
Technical Coordinator*

Administered the Snowflake cloud data warehouse including schema design under the Medallion Architecture and query optimization. Maintained version-controlled code repositories and coordinated technical documentation standards. Contributed to the React-based frontend development.

AmatAlrahman Sayed

*Data Preparation &
Simulation Engineer*

Owned the data preprocessing and quality assurance lifecycle including deduplication, normalization, and validation pipelines. Engineered a real-time prescription simulation engine using a Python Kafka Producer to emulate live pharmacy streams, enabling rigorous pipeline stress testing and end-to-end validation.

Ali Gad

*Big Data & Pipeline
Orchestration Engineer*

Developed distributed data processing workflows using Databricks (Apache Spark/PySpark) and designed fault-tolerant, automated Airflow DAGs to ensure reliable, continuous data integration from ingestion through warehousing.

03 Project Objectives

01 Real-Time Processing

Construct a high-throughput pipeline capable of receiving, validating, and processing medical prescriptions end-to-end in under three seconds — from pharmacy input to Neo4j interaction response.

02 AI-Driven Drug Name Standardization

Deploy Groq LLMs to automatically cleanse and normalize free-text commercial drug names with a target AI confidence score of 90% or above, eliminating manual correction overhead.

03 Medical Safety & Interaction Detection

Substantially reduce medication errors by enabling instant graph-traversal-based detection of drug-drug interactions, N-degree polypharmacy risks, and patient-specific contraindications.

04 Decision Support Dashboards

Deliver dynamic, near-real-time Power BI dashboards monitoring drug dispensing patterns, interaction hotspots, and system health metrics for institutional administrators and clinical teams.

05 Personalized Medicine & EHR Integration

Integrate each patient's complete medical history — chronic conditions, allergies, and prior adverse events — into the graph engine to generate context-aware, patient-specific medication alerts.

06 Containerized Deployment

Deliver the entire preprocessing pipeline and web application as Docker containers, ensuring reproducible, portable, and production-ready deployments across any cloud or on-premises environment.

04 Tools & Technologies

DATA STREAMING

Tool / Component	Description
Apache Kafka (Aiven Cloud)	Distributed event streaming platform. Prescription events published to dedicated Kafka topics and consumed by PySpark Streaming consumers in real time, with replay, partitioning, and consumer group fault tolerance.

DATA PROCESSING & ORCHESTRATION

Tool / Component	Description
Apache Spark / PySpark	Distributed batch and micro-batch processing on Databricks. Handles transformations, graph lookup enrichment, and Snowflake loading.
Apache Airflow	Schedules all batch DAGs, incremental Snowflake updates, Neo4j bulk refresh tasks, and monitoring alert pipelines with retry and email alerting.
Python 3.x	Core scripting language for data cleaning scripts, Kafka Producer simulation engine, Groq API integration utilities, and ETL helper modules.

AI & MACHINE LEARNING

Tool / Component	Description
Groq API — LLM	Drug name standardization and entity extraction. Maps free-text commercial names to canonical INN identifiers with >90% confidence via prompt-engineered inference.
GraphRAG Architecture	Merges LLM semantic reasoning with Neo4j graph retrieval to power the Virtual Medical Assistant for natural-language clinical queries.

GRAPH DATABASE

Tool / Component	Description
Neo4j Aura Cloud	Property graph modeling drugs, conditions, interactions, contraindications, patient profiles, and ATC classifications. Enables N-degree path traversal, reverse symptom tracing, and safe alternative suggestions.

CLOUD DATA WAREHOUSE

Tool / Component	Description
Snowflake	Cloud-native DWH using Medallion Architecture (Bronze / Silver / Gold layers) and OBT schema. Ingested via Snowpipe; queried by Power BI for executive and clinical dashboards.

BACKEND API

Tool / Component	Description
Node.js + Express.js	RESTful API server for prescription submission, interaction results, patient history lookup, and AI assistant queries. JWT authentication and role-based access control for Doctors, Pharmacists, and Admins.
Socket.io (WebSocket)	Real-time push notifications to the frontend when a critical drug interaction alert is triggered — ensuring zero-latency clinical warning delivery.

FRONTEND

Tool / Component	Description
React.js	Single-page application for prescription input, real-time alert display, patient profile management, and Power BI dashboard embedding. Communicates via REST and WebSocket.

CONTAINERIZATION

Tool / Component	Description
Docker & Docker Compose	Full containerization of all services — preprocessor, API, frontend, Airflow, and Neo4j. Docker Compose orchestrates startup, networking, volumes, and health checks for consistent deployments.

BI & VISUALIZATION

Tool / Component	Description
Power BI	Interactive dashboards for drug dispensing trends, interaction hotspot maps, AI confidence metrics, and pipeline health KPIs. Connected directly to Snowflake Gold layer.

0 5 Milestones & Delivery Schedule

M 01	M 02	M 03	M 04
Infrastructure & Databases	Processing & AI Integration	Pipeline, Docker & Backend	Frontend & Final Delivery
12-15 Days	12-15 Days	12-15 Days	12-15 Days

Milestone 01 Infrastructure & Database Setup 12-15 Days

- Provision Apache Kafka cluster on Aiven Cloud; configure topics (prescriptions-raw, prescriptions-processed, alerts-output) with replication factors and retention policies.
- Design and deploy the Neo4j Aura Cloud graph schema: nodes (Drug, Condition, Patient, Ingredient, ATC_Class) and relationships (INTERACTS_WITH, CONTRAINDICATED_FOR, HAS_CONDITION, HAS_ALLERGY, BELONGS_TO_CLASS) with indexed properties.
- Seed the Neo4j graph with 2,000+ drug entities, 500+ drug-drug interaction pairs, and ATC classification hierarchy sourced from DrugBank and OpenFDA.
- Initialize Snowflake environment: create Bronze, Silver, and Gold schemas; configure Snowpipe for continuous ingestion from the processed Kafka topic.
- Set up Apache Airflow; configure connections to Kafka, Neo4j, and Snowflake; verify end-to-end connectivity across all services.
- Establish Docker Compose baseline: define service containers for Kafka consumer, Airflow scheduler, Neo4j seed scripts, and Snowflake loaders.
- Deliverable: Live, connected infrastructure stack with verified Kafka → Neo4j → Snowflake data flow on a synthetic test prescription.

Milestone 02 Data Processing & AI Integration 12-15 Days

- Develop the Python Kafka Producer simulation engine to emulate realistic pharmacy prescription streams at 50–200 events/minute for load testing.
- Build the Groq LLM integration module: prompt engineering for drug name normalization, confidence scoring, fallback strategies for low-confidence results, and caching for repeated lookups.
- Implement the PySpark Streaming consumer: consume raw prescription events, invoke the Groq normalization service, validate required fields, and produce enriched events to the processed topic.
- Develop the Neo4j Cypher query library: pairwise interaction detection, N-degree polypharmacy aggregation, safe alternative engine (same ATC class, no interaction path), and reverse symptom traversal.
- Build the GraphRAG pipeline: hybrid retrieval combining Neo4j graph context extraction with Groq LLM generation for the Virtual Medical Assistant.

- Validate AI accuracy against a 200-prescription benchmark dataset; tune prompts until $\geq 90\%$ confidence score is consistently achieved.
- Deliverable: Fully operational AI cleaning and graph interaction detection pipeline with documented accuracy metrics.

Milestone 03 Pipeline, Warehousing, Dockerization & Backend 12–15 Days

- Finalize PySpark batch processing jobs for end-of-day prescription aggregation, KPI computation, and Gold layer population in Snowflake.
- Author all production Airflow DAGs: daily prescription batch, Neo4j bulk interaction refresh, Snowflake table maintenance, and data quality monitoring — with retry logic and email alerting.
- Design and implement the Node.js (Express) RESTful API: prescription submission, interaction query, patient history, AI assistant, and admin dashboard data endpoints.
- Implement JWT authentication middleware, role-based access control (Doctor / Pharmacist / Admin), request validation, and rate limiting.
- Add Socket.io WebSocket layer for real-time alert push to connected frontend clients.
- Containerize all services with Docker: write Dockerfiles for the preprocessor, API server, frontend, and Airflow. Author docker-compose.yml with health checks, volume mounts, and .env-based configuration.
- Deliverable: Fully containerized, integration-tested pipeline and API stack deployable via a single docker-compose up command.

Milestone 04 Frontend, Visualization & Final Delivery 12–15 Days

- Develop the React.js web application: prescription input form, real-time interaction alert panel with severity colour-coding, patient history view, AI assistant chat interface, and admin dashboard page.
- Integrate frontend with Node.js API via REST and Socket.io for live alert push notifications.
- Build Power BI dashboards: drug dispensing trends, interaction hotspot maps, AI confidence score trends, Airflow pipeline health KPIs, and near-real-time prescription volume gauge (≤ 10 sec lag).
- Containerize the React app in an Nginx production build Docker image; finalize the complete docker-compose stack.
- Perform end-to-end user acceptance testing: simulate a doctor submitting a polypharmacy prescription and verify alerts, safe alternatives, and dashboard update within defined SLAs.
- Prepare the final technical report: architecture diagrams, API documentation, database schema docs, Docker deployment guide, and performance benchmark results.
- Deliverable: Live web application, published Power BI dashboards, complete technical documentation, and defense-ready presentation.

06 Key Performance Indicators

The following KPIs are customized to reflect the advanced architecture of DataDose. Each metric corresponds to a specific technical layer and is directly traceable to a project objective.

DATA PREPROCESSING & AI QUALITY

$\geq 90\%$

AI Confidence Score in Drug Name
Standardization

100%

Automated Handling of Missing &
Duplicate Data

≤ 2 sec

End-to-End Prescription Latency per
Transaction

GRAPH & WAREHOUSE INTEGRATION

$\geq 95\%$

Cypher Query Accuracy vs. Medical
Benchmarks

< 5 sec

Avg. Snowflake Merge & Neo4j Query
Time / Batch

100%

Airflow DAG Uptime — Zero Pipeline
Failures

VISUALIZATION & DASHBOARD

< 3 sec

Power BI Dashboard Load Time

≤ 10 sec

Data Refresh Lag: Input to Dashboard

$> 90\%$

Required Metrics Visually Represented

CONTAINERIZATION & DEPLOYMENT

1 Command

Full Stack Start via docker-compose up

< 3 min

Cold-Start Time for Complete Docker Stack

Presentation Standard: 100% completion of the final technical report with all source code, architecture diagrams, and deployment documentation.

07 Killer Features & Competitive Advantage

DataDose's decisive competitive advantage is its transformation from a passive **risk-alert** model into an active **solution-providing** model — powered by five exclusive capabilities built on the combined strength of Neo4j graph traversal and large language model reasoning. None of these features exist in combination in any commercially available pharmacy management system today.

01 Smart Safe Alternatives Engine

Problem

Existing clinical decision support systems warn of an interaction between Drug A and Drug B but provide no actionable path forward, leaving the prescribing physician to manually search for alternatives under time pressure.

Solution

Upon detecting a conflict, the Neo4j engine traverses the ATC classification graph to identify a pharmacologically equivalent Drug C — one that treats the same condition and presents no INTERACTS_WITH relationship with any current medication — returning a safe, actionable alternative within milliseconds.

02 N-Degree Polypharmacy Analysis

Problem

Traditional pairwise comparison tools evaluate drugs in isolation. They systematically fail patients on complex poly-medication regimens where three or more drugs share a cumulative adverse effect pathway — a risk entirely invisible in pairwise evaluation.

Solution

DataDose models the entire prescription as an integrated network. If three or more drugs share convergent edges toward the same adverse-effect node, the system aggregates these pathways and issues a calibrated cumulative alert — e.g., '3x multiplied risk of severe hypotension' — with supporting pharmacological evidence.

03 Reverse Symptom Tracer

Problem

When a patient presents an unexpected adverse event, the clinician must manually review prescribing information for every drug in the patient's regimen — a process that can take hours in complex cases.

Solution

DataDose accepts a reported side effect as input and executes a reverse graph traversal from the adverse-effect node, propagating backward through CAUSES relationships to identify the prime suspect drug within the patient's current medication list — reducing root-cause analysis from hours to seconds.

04 GraphRAG AI Medical Assistant

Problem

Traditional keyword search and static drug reference databases cannot answer complex, context-sensitive clinical questions requiring reasoning across multiple pharmacological relationships simultaneously.

Solution

By fusing Groq LLM semantic reasoning with Neo4j graph retrieval (GraphRAG), DataDose answers natural-language queries such as: 'A diabetic patient on Metformin with hypertension — what cold medication is safe?' The system retrieves graph context, constructs a structured prompt, and returns a safe, evidence-backed recommendation.

05 Patient-Centric History Integration

Problem

Conventional systems evaluate drug-to-drug interactions in isolation, ignoring the patient's individual clinical profile. Two drugs may be pharmacologically compatible in general yet pose an imminent threat to a patient with a specific chronic condition or documented allergy.

Solution

DataDose integrates the patient's full clinical profile as first-class graph nodes: (Patient)-[HAS_CONDITION]->(Condition). Upon prescription entry, every new drug is evaluated against all CONTRAINDICATED_FOR relationships in the patient's history, issuing personalized alerts such as: 'Warning: This medication is contraindicated for the gastric ulcer recorded in this patient's history since 2024.'

DataDose

This graduation project was submitted to the Ministry of Communications and Information Technology as part of the **Digital Egypt Pioneers Initiative (DEPI)**, in collaboration with **eYouth Academy**. The project demonstrates the implementation of enterprise-grade **data engineering**, **cloud infrastructure**, and **applied artificial intelligence** to enhance **patient safety and healthcare decision-making**.